

● MEDIUM

● INFORMATION AND IDEAS

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12

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

The Indus River valley civilization flourished in South Asia from 3300 BCE to 1300 BCE. Many examples of the civilization's writing system exist, but researchers haven't yet deciphered it or identified which ancient language it represents. Nevertheless, archaeologists have found historical artifacts, such as clay figures and jewelry, that provide information about the civilization's customs and how its communities were organized. The archaeologists' findings therefore suggest that _____blank

Which choice most logically completes the text?

- A. investigating an ancient civilization is easier without knowledge of the civilization's language.
- B. knowing an ancient civilization's language isn't necessary in order to learn details about the civilization.
- C. archaeological research should focus on finding additional artifacts rather than deciphering ancient languages.
- D. examining the civilization's historical artifacts has resolved the debate about this civilization's language.

“The Yellow Wallpaper” is an 1892 short story by Charlotte Perkins Gilman. In the story, the narrator expresses mixed feelings about her surroundings: _____ blank

Which quotation from “The Yellow Wallpaper” most effectively illustrates the claim?

- A. “This wallpaper has a kind of sub-pattern in a different shade, a particularly irritating one, for you can only see it in certain lights, and not clearly then.”
- B. “By moonlight—the moon shines in all night when there is a moon—I wouldn’t know it was the same paper.”
- C. “I’m really getting quite fond of the big room, all but that horrid [wall]paper.”
- D. “The color is repellant, almost revolting; a smouldering, unclean yellow, strangely faded by the slow-turning sunlight.”

Biologists have predicted that birds' feather structures vary with habitat temperature, but this hadn't been tested in mountain environments. Ornithologist Sahas Barve studied feathers from 249 songbird species inhabiting different elevations—and thus experiencing different temperatures—in the Himalaya Mountains. He found that feathers of high-elevation species not only have a greater proportion of warming downy sections to flat and smooth sections than do feathers of low-elevation species, but high-elevation species' feathers also tend to be longer, providing a thicker layer of insulation.

Which choice best states the main idea of the text?

- A. Barve's investigation shows that some species of Himalayan songbirds have evolved feathers that better regulate body temperature than do the feathers of other species, contradicting previous predictions.
- B. Barve found an association between habitat temperature and feather structure among Himalayan songbirds, lending new support to a general prediction.
- C. Barve discovered that songbirds have adapted to their environment by growing feathers without flat and smooth sections, complicating an earlier hypothesis.
- D. The results of Barve's study suggest that the ability of birds to withstand cold temperatures is determined more strongly by feather length than feather structure, challenging an established belief.

The Cretaceous pterosaur *Tupandactylus navigans* is known for having an anomalously oversized head crest. Until an almost complete fossil skeleton was found in Brazil, paleontologists had been able to study only skull specimens from *T. navigans*, though it was presumed that, like other pterosaurs, the species's primary form of locomotion was powered flight. Examining the fuller skeleton in 2016, Victor Beccari and his team determined that *T. navigans* had long hind legs, short wings, and an unusually long neck—characteristics that, combined with the creature's large-crested head, would have made sustained flight difficult and walking upright relatively comfortable. Based on these findings the team suggests that *T. navigans* likely _____ blank

Which choice most logically completes the text?

- A. flew for longer distances than did other pterosaur species that had oversized head crests.
- B. had longer wings than other pterosaur species considered to have been comfortable walking.
- C. had a smaller head than researchers expected based on the earlier *T. navigans* skull specimens.
- D. flew for shorter distances and spent more time walking than researchers previously thought.

External shopping cues are a type of marketing that uses obvious messaging—a display featuring a new product, for example, or a “buy one, get one free” offer—to entice consumers to make spontaneous purchases. In a study, data scientist Sam K. Hui and colleagues found that this effect can also be achieved with a less obvious cue: rearranging a store’s layout. The researchers explain that trying to find items in new locations causes shoppers to move through more of the store, exposing them to more products and increasing the likelihood that they’ll buy an item they hadn’t planned on purchasing.

Which response from a survey given to shoppers who made a purchase at a retail store best supports the researchers’ explanation?

- A. “I needed to buy some cleaning supplies, but they weren’t in their regular place. While I was looking for them, I saw this interesting notebook and decided to buy it, too.”
- B. “I didn’t buy everything on my shopping list today. I couldn’t find a couple of the items in the store, even though I looked all over for them.”
- C. “The store sent me a coupon for a new brand of soup, so I came here to find out what kinds of soup that brand offers. I decided to buy a few cans because I had the coupon.”
- D. “This store is larger than one that’s closer to where I live, and it carries more products. I came here to buy some things that the other store doesn’t always have.”

Philadelphia's Black Pearl Chamber Orchestra, founded by Jeri Lynne Johnson, performs classical music, from well-known compositions by Beethoven to contemporary works by Jessie Montgomery. For the orchestra's iConduct! program, Johnson invites community members to learn some basic elements of conducting and then experience conducting the Black Pearl orchestra themselves.

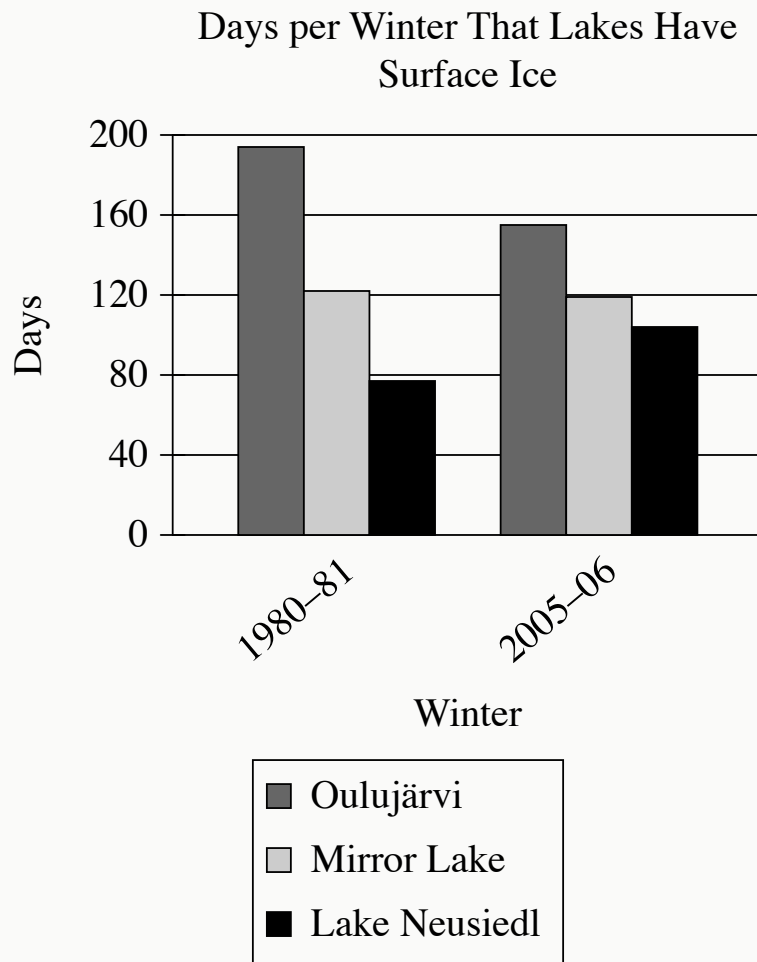
Which choice best states the main idea of the text?

- A. The Black Pearl orchestra performs music from all over the world but mostly performs music composed by Philadelphians.
- B. Johnson founded the Black Pearl orchestra to perform classical music by contemporary artist Jessie Montgomery.
- C. The Black Pearl orchestra gives community members the chance to both listen to and participate in classical music performance.
- D. Johnson has community members conduct an orchestra to demonstrate how difficult the task is.

Biologist Natacha Bodenhausen and colleagues analyzed the naturally occurring bacterial communities associated with leaves and roots of wild *Arabidopsis thaliana*, a small flowering plant. The researchers found many of the same bacterial genera in both the plants' leaves and roots. To explain this, the researchers pointed to the general proximity of *A. thaliana* leaves to the ground and noted that rain splashing off soil could bring soil-based bacteria into contact with the leaves. Alternatively, the researchers noted that wind, which may be a source of bacteria in the aboveground portion of plants, could also bring bacteria to the soil and roots. Either explanation suggests that _____ blank

Which choice most logically completes the text?

- A. bacteria carried by wind are typically less beneficial to *A. thaliana* than soil-based bacteria are.
- B. some bacteria in *A. thaliana* leaves and roots may share a common source.
- C. many bacteria in *A. thaliana* leaves may have been deposited by means other than rain.
- D. *A. thaliana* leaves and roots are especially vulnerable to harmful bacteria.



- For each data category, the following bars are shown:
 - Oulujärvi
 - Mirror Lake
 - Lake Neusiedl
- The data for the 2 categories are as follows:
 - 1980-81:
 - Oulujärvi: 194 days
 - Mirror Lake: 122 days
 - Lake Neusiedl: 77 days
 - 2005-06:
 - Oulujärvi: 155 days

- Mirror Lake: 119 days
- Lake Neusiedl: 104 days

It is common for freshwater lakes near or above a latitude of 45° north of the equator, like Lake Mjøsa in Norway, to accumulate surface ice in winter. The amount and duration of ice depends on many factors, including local weather conditions as well as the lake's depth, volume, and surface area, but a climate researcher claims that some lakes in these latitudes have seen a decline in the duration of ice between the early 1980s and the mid-2000s. She cites as a typical example _____blank

Which choice most effectively uses data from the graph to complete the researcher's example?

- A. both Lake Neusiedl and Oulujärvi, which had fewer than 195 days of ice in the winter of 1980–81.
- B. Lake Neusiedl, which had more days of ice in the winter of 2005–06 than it did in the winter of 1980–81.
- C. Oulujärvi, which had fewer days of ice in the winter of 2005–06 than it did in the winter of 1980–81.
- D. both Lake Neusiedl and Oulujärvi, which had more than 105 days of ice in the winter of 2005–06.

In a study by Mika R. Moran, Daniel A. Rodriguez, and colleagues, residents of Quito, Ecuador, and Lima, Peru, were surveyed about parks in their cities. Of the 618 respondents from Quito, 82.9% indicated that they use the city's parks, and of the 663 respondents from Lima, 72.7% indicated using city parks. Given that the percentage of Quito respondents who reported living within a 10-minute walk of a park was much lower than that reported by Lima respondents, greater proximity alone can't explain the difference in park use.

The text makes which point about the difference between the proportions of Quito residents and Lima residents using parks?

- A. It was much larger than the researchers conducting the study expected.
- B. It is caused by something other than the parks' proximity to city residents.
- C. It could be due to inaccuracies in the survey results.
- D. It was calculated using sources that predate the survey.

Euphorbia esula (leafy spurge) is a Eurasian plant that has become invasive in North America, where it displaces native vegetation and sickens cattle. *E. esula* can be controlled with chemical herbicides, but that approach can also kill harmless plants nearby. Recent research on introducing engineered DNA into plant species to inhibit their reproduction may offer a path toward exclusively targeting *E. esula*, consequently _____ blank

Which choice most logically completes the text?

- A. making individual *E. esula* plants more susceptible to existing chemical herbicides.
- B. enhancing the ecological benefits of *E. esula* in North America.
- C. enabling cattle to consume *E. esula* without becoming sick.
- D. reducing invasive *E. esula* numbers without harming other organisms.